



Name: Cohort/Batch: Date: Score:

SPRING BOOT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A covering auto-config, JPA, security, and testing

TOTAL: 10 QUESTIONS

Q1 What is Spring Boot and how does it differ from the Spring Framework?

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Q6 How do you configure a relational database in Spring Boot?

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Q2 How does Spring Boot auto-configuration work under the hood?

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Q7 How do you replace the default embedded Tomcat with Jetty?

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Q3 What does @SpringBootApplication combine and why is each part needed?

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Q8 How do you implement global REST exception handling in Spring Boot?

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Q4 How does Spring Boot externalize and prioritize configuration?

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Q9 What are Spring Boot profiles and how do you activate one?

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Q5 What is Spring Boot Actuator and what does it expose?

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Q10 How do you write a Spring Boot integration test?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Spring Boot is a Spring module that

Mitigation

Spring Boot is a Spring module that provides auto-configuration, embedded servers, and starter POMs. Unlike plain Spring, it needs zero XML and runs as a self-contained JAR, removing the need to deploy to a separate app server.

A6 Add spring-boot-starter-data-jpa and a JDBC driver. See

Servability

Add spring-boot-starter-data-jpa and a JDBC driver. Set spring.datasource.url, username, and password in application.properties. Boot auto-wires a DataSource, EntityManagerFactory, and JpaTransactionManager without any explicit @Bean definitions.

A2 @EnableAutoConfiguration loads candidate classes from

mitives

@EnableAutoConfiguration loads candidate classes from AutoConfiguration.imports. Each class uses @ConditionalOnClass or @ConditionalOnMissingBean to activate only when specific dependencies or beans are present on the classpath.

A7 Exclude spring-boot-starter-tomcat from the web starter, then

then

Exclude spring-boot-starter-tomcat from the web starter, then add spring-boot-starter-jetty. Auto-configuration detects Jetty on the classpath and selects it. Undertow is a third option available the same way.

A3 It merges @Configuration (declares beans),

Hardening

It merges @Configuration (declares beans), @EnableAutoConfiguration (activates Boot's auto-config machinery), and @ComponentScan (recursively discovers @Component, @Service, @Repository in the package tree).

A8 @RestControllerAdvice marks a class as a cross-cutting

performance

@RestControllerAdvice marks a class as a cross-cutting error handler. Methods annotated @ExceptionHandler(SomeEx.class) intercept matched exceptions and return a ResponseEntity with a structured error body and HTTP status code.

A4 Boot loads sources in ascending-priority order:

Gotchas

Boot loads sources in ascending-priority order: application.yml, profile files, OS environment variables, and CLI args each override the previous. @ConfigurationProperties binds a prefix to a typed POJO with IDE-validation support.

A9 Profiles separate environment configuration.

Assessment

Profiles separate environment configuration. application-prod.yml is only loaded when the prod profile is active. Activate via spring.profiles.active=prod in properties, the SPRING_PROFILES_ACTIVE env var, or --spring.profiles.active=prod CLI arg.

A5 Actuator adds management HTTP endpoints: /actuator/health

health

Actuator adds management HTTP endpoints: /actuator/health (liveness/readiness), /actuator/metrics (counters, timers), /actuator/loggers (change log levels at runtime), and /actuator/env. Each endpoint is securable and toggleable.

A10 @SpringBootTest starts a full application context. Add

optimization

@SpringBootTest starts a full application context. Add @AutoConfigureMockMvc to inject a MockMvc for HTTP-level assertions without a real server. Use @MockBean to replace infrastructure beans (e.g., database, email) with Mockito doubles.